Alexandra Medina

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Portfolio: https://a-med24.github.io/

EDUCATION

University of Illinois at Chicago Chicago, IL
Bachelor of Science, Computer Science and Statistics Expected May 2028
3.5 GPA

SKILLS

Programming & Tools: Python, C/C++, Java, JavaScript, SQL, R, Go, React, Microsoft Azure, PowerShell, Raspberry Pi, TCP/IP Networking, Git
Other Skills: Customer Service, Mentorship, Multimedia Editing
Certifications: Microsoft 365 Fundamentals, WordPress 101

COURSEWORK

Systems Programming, Machine Organization, Computer Design, Data Structures, Software Design, Programming Practicum, Coding for Civic Data Applications, Languages and Automata

WORK EXPERIENCE

Argonne National Laboratory – Q-NEXT May 2026 – June 2026
Contingent Worker | Quantum AI/ML Intern

- Developed a Python-based control framework for an autonomous robotic arm used in quantum materials research, enabling reliable TCP/IP communication, and remote operation
- Built a command-line interface to manually control the robotic system, teach and store positions, and integrate logging for reproducible automation workflow
- Created automated unit tests using mock TCP sockets to validate networking, collision detection, interlock logic, and watchdog functionality without requiring physical hardware

University of Illinois at Chicago – Endpoint Management September 2025 – PRESENT
Computer Specialist I

- Provide second-tier technical support for campus-wide systems
- Develop and deploy Powershell automation scripts in Microsoft Azure virtual environments
- Reimage and configure enterprise PCs using OSDCloud, Hiren tool, and JAMF

PROJECTS

IBM Bob AI Build Hackathon: Fatigue Console August 2026

- Built the backend utilizing Python, including a mock mission database, data pipeline, What-If Simulator, and dependency graph to model task-reassignment scenarios and crew workload/fatigue tradeoffs
- Integrated outputs with IBM Granite to generate explainable Mission Decision Briefs from structured mission data, surfacing fatigue drivers, mission implications, intervention tradeoffs, and recommended actions

SparkHacks 2026: LitQuest February 2026

- Built a full-stack web application and app prototype in 24 hours that transforms classic literature into interactive visual novels using React and Figma
- Designed and integrated a MongoDB database to render dialogue and narrative progression

EXTRACURRICULAR

Capture the Flame – Experience Board Member September 2026 - PRESENT

- Develop hands-on security challenges that test participants' technical problem-solving and cybersecurity skills with a 10-member team
- Contribute to challenge testing, solution validation, and documentation